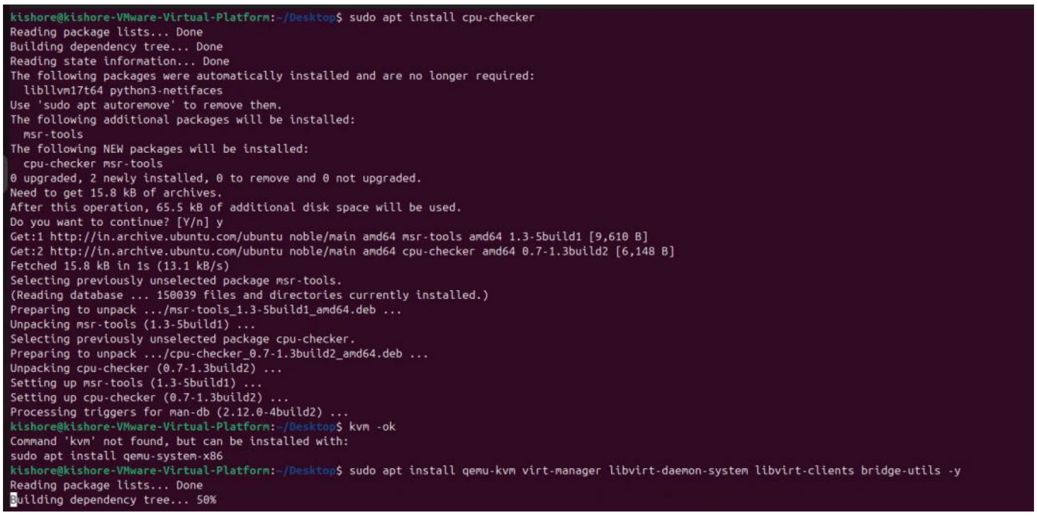
Assignment No. 2

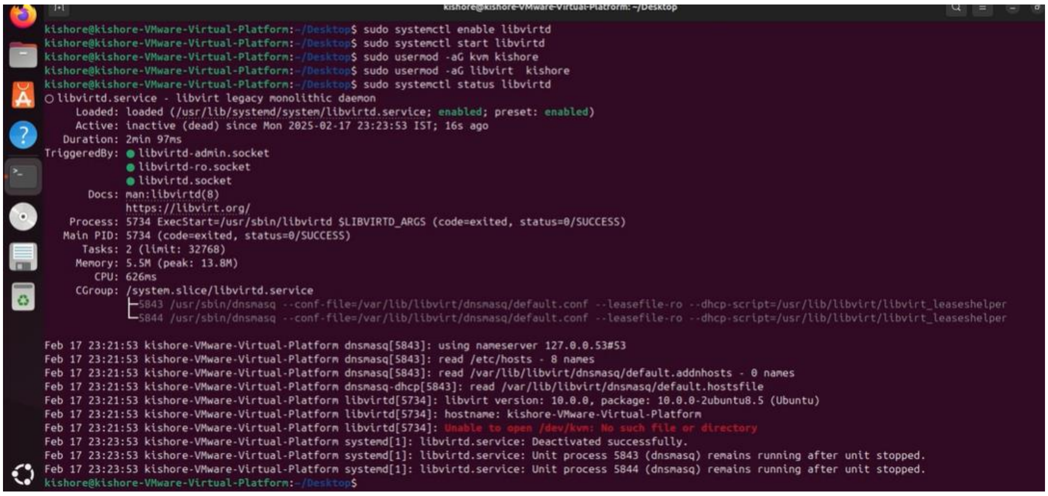
Procedure

Procedure (Step-by-Step)

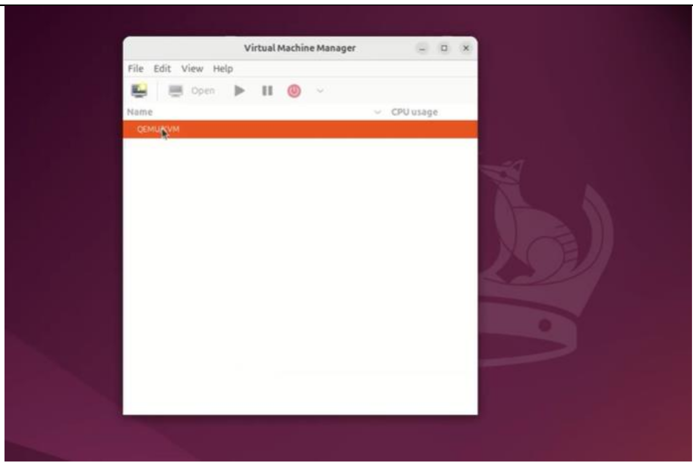
- Step 1: Update System
- Step 2: Install KVM and Required Packages
- Step 3: Verify Installation



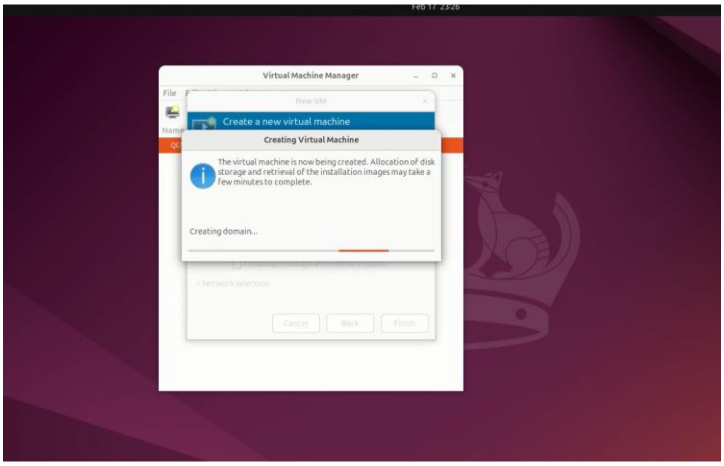
- Step 4: Start and Enable Services
- Step 5: Add User to libvirt Group



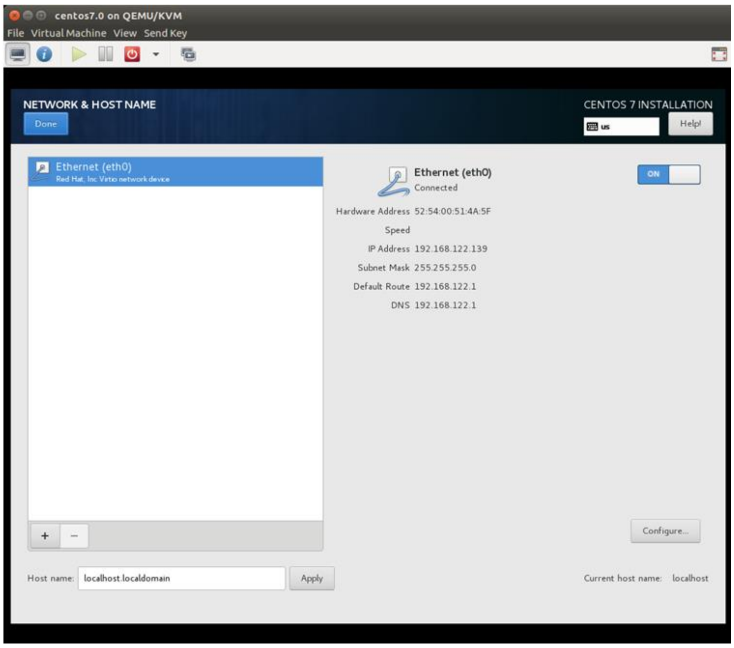
- Step 6: Launch Virtual Machine Manager



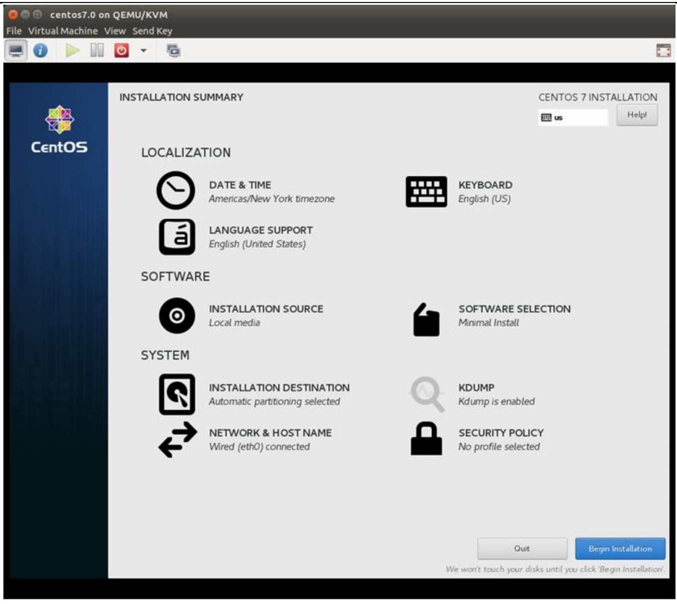
- Step 7: Create a Virtual Machine



- Step 8: Configure Networking



- Step 9: Manage Virtual Machine



Observations and Results

- The system successfully detected hardware virtualization support.
- KVM packages were installed without errors.
- The kvm-ok command confirmed that KVM is enabled.
- The libvirtd service was running successfully.
- Virtual Machine Manager launched correctly.
- A virtual machine was created and the OS installation completed successfully.
- The VM was able to boot and run independently.

Analysis

The practical shows how Kernel-based Virtual Machine uses hardware virtualization to run multiple virtual machines efficiently on a single system. It provides high performance with low overhead due to its integration with the Linux kernel. Overall, it is a scalable and cost-effective virtualization solution.

Conclusion

The practical demonstrated the installation and configuration of KVM as a virtualization solution. It showed how a Linux system can be converted into a hypervisor to run multiple virtual machines efficiently. KVM provides a powerful, open-source, and scalable virtualization platform suitable for cloud and enterprise environments.